

Final-term

Time: Jun. 9 Monday, 1:30-3:30pm

Location: 教学中心101

Covers Chapter 8~13

Format: 10 selection questions and 4 problems; closed-book, one A4-size cheat sheet allowed

Grade: %35 of the total grade

- 计算器 
- 涂卡笔 

Final-term review

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Disclaimer

- Topics covered in this review may not appear in the exam.
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NP and Computational Intractability

Key concepts

- Decision problems, certifiers and certificates
- P, NP, co-NP, EXP, PSPACE, NP-complete, NP-hard
- Polynomial transformation and polynomial-time reduction

Establish NP-completeness of problem Y.

- Step 1. Show that Y is in NP.
- Step 2. Choose an NP-complete problem X.
- Step 3. Prove that $X \leq_p Y$.

Basic reduction strategies.

- Reduction by simple equivalence.
- Reduction from special case to general case.
- Reduction by encoding with gadgets.

Discussion on P versus NP

NP and Computational Intractability

Some typical hard problems

- Independent set, vertex cover
- SAT, 3-SAT
- Hamiltonian cycle, matching and 3D-matching, Partition (introduced in Appr. Alg. part), SUBSET-SUM, KNAPSACK

PSPACE

PSPACE. Decision problems solvable in polynomial **space**.

Theorem. $NP \subseteq PSPACE \subseteq EXPTIME$

PSPACE-Complete. Problem Y is PSPACE-complete if (i) Y is in PSPACE and (ii) for every problem X in PSPACE, $X \leq_p Y$.

PSPACE-Complete problems

- QSAT
- Competitive Facility Location

Coping With NP-Completeness

Must sacrifice one of three desired features.

- Solve problem to optimality.
- Solve problem in polynomial time.
- Solve arbitrary instances of the problem.

Randomized Algorithms

Basic math. Linearity of expectation.

Monte Carlo algorithm. Guaranteed running time, likely to find correct answer.

- Contraction algorithm for global min cut

Las Vegas algorithm. Guaranteed to find correct answer, likely to run in certain time.

- Johnson's MAX-3SAT algorithm
- Randomized quicksort

Transformation

- Can always convert a Las Vegas algorithm into Monte Carlo, but no known method to convert the other way.

RP and ZPP

- Definition of RP and ZPP.
- $P \subseteq ZPP \subseteq RP \subseteq NP$.

Approximation Algorithms

ρ -approximation algorithm.

- Guaranteed to run in poly-time.
- Guaranteed to solve arbitrary instance of the problem
- Guaranteed to find solution within ratio ρ of true optimum.

Example problems

- Load Balancing
 - Greedy algorithm is a $3/2$ -approximation
- Center Selection
 - Greedy algorithm is a 2-approximation
- Weighted Vertex Cover
 - Pricing method is a 2-approximation
 - Linear programming + rounding is a 2-approximation
- Knapsack Problem
 - Polynomial Time Approximation Scheme: $(1 + \varepsilon)$ -approximation algorithm for any constant $\varepsilon > 0$
 - Rounding and scaling

Good luck!